

INTERNATIONAL STANDARD

IEC
60384-2-1

QC 300401

Second edition
2005-11

Fixed capacitors for use in electronic equipment –

Part 2-1:

Blank detail specification:

Fixed metallized polyethylene-terephthalate film dielectric d.c. capacitors – Assessment levels E and EZ



Reference number
IEC 60384-2-1:2005(E)

Publication numbering

As from 1 January 1997 all IEC publications are issued with a designation in the 60000 series. For example, IEC 34-1 is now referred to as IEC 60034-1.

Consolidated editions

The IEC is now publishing consolidated versions of its publications. For example, edition numbers 1.0, 1.1 and 1.2 refer, respectively, to the base publication, the base publication incorporating amendment 1 and the base publication incorporating amendments 1 and 2.

Further information on IEC publications

The technical content of IEC publications is kept under constant review by the IEC, thus ensuring that the content reflects current technology. Information relating to this publication, including its validity, is available in the IEC Catalogue of publications (see below) in addition to new editions, amendments and corrigenda. Information on the subjects under consideration and work in progress undertaken by the technical committee which has prepared this publication, as well as the list of publications issued, is also available from the following:

- **IEC Web Site** (www.iec.ch)

- **Catalogue of IEC publications**

The on-line catalogue on the IEC web site (www.iec.ch/searchpub) enables you to search by a variety of criteria including text searches, technical committees and date of publication. On-line information is also available on recently issued publications, withdrawn and replaced publications, as well as corrigenda.

- **IEC Just Published**

This summary of recently issued publications (www.iec.ch/online_news/justpub) is also available by email. Please contact the Customer Service Centre (see below) for further information.

- **Customer Service Centre**

If you have any questions regarding this publication or need further assistance, please contact the Customer Service Centre:

Email: custserv@iec.ch
Tel: +41 22 919 02 11
Fax: +41 22 919 03 00

INTERNATIONAL STANDARD

IEC 60384-2-1

QC 300401

Second edition
2005-11

Fixed capacitors for use in electronic equipment –

Part 2-1:

Blank detail specification:

Fixed metallized polyethylene-terephthalate film dielectric d.c. capacitors – Assessment levels E and EZ

© IEC 2005 — Copyright - all rights reserved

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from the publisher.

International Electrotechnical Commission, 3, rue de Varembe, PO Box 131, CH-1211 Geneva 20, Switzerland
Telephone: +41 22 919 02 11 Telefax: +41 22 919 03 00 E-mail: inmail@iec.ch Web: www.iec.ch



Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

PRICE CODE

M

For price, see current catalogue

INTERNATIONAL ELECTROTECHNICAL COMMISSION

FIXED CAPACITORS FOR USE IN ELECTRONIC EQUIPMENT –**Part 2-1: Blank detail specification: Fixed metallized
polyethylene-terephthalate film dielectric d.c. capacitors –
Assessment levels E and EZ**

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with an IEC Publication.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 60384-2-1 has been prepared by IEC technical committee 40: Capacitors and resistors for electronic equipment.

This second edition cancels and replaces the first edition published in 1982 and amendment 1 published in 1987 and constitutes minor revisions related to tables, figures and references.

The text of this standard is based on the following documents:

FDIS	Report on voting
40/1594/FDIS	40/1627/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

IEC 60384 consists of the following parts, under the (new) general title *Fixed capacitors for use in electronic equipment*:

- Part 1: Generic specification
- Part 2: Sectional specification: Fixed metallized polyethylene-terephthalate film dielectric d.c. capacitors
- Part 3: Sectional specification: Fixed tantalum chip capacitors
- Part 4: Sectional specification: Aluminium electrolytic capacitors with solid and non-solid electrolyte
- Part 5: Sectional specification: Fixed mica dielectric d.c. capacitors with a rated voltage not exceeding 3000 V – Selection of methods of test and general requirements
- Part 6: Sectional specification: Fixed metallized polycarbonate film dielectric d.c. capacitors
- Part 7: Sectional specification: Fixed polystyrene film dielectric metal foil d.c. capacitors
- Part 8: Sectional specification: Fixed capacitors of ceramic dielectric, Class 1
- Part 9: Sectional specification: Fixed capacitors of ceramic dielectric, Class 2
- Part 11: Sectional specification: Fixed polyethylene-terephthalate film dielectric metal foil d.c. capacitors
- Part 12: Sectional specification: Fixed polycarbonate film dielectric metal foil d.c. capacitors
- Part 13: Sectional specification: Fixed polypropylene film dielectric metal foil d.c. capacitors
- Part 14: Sectional specification: Fixed capacitors for electromagnetic interference suppression and connection to the supply mains
- Part 15: Sectional specification: Fixed tantalum capacitors with non-solid or solid electrolyte
- Part 16: Sectional specification: Fixed metallized polypropylene film dielectric d.c. capacitors
- Part 17: Sectional specification: Fixed metallized polypropylene film dielectric a.c. and pulse capacitors
- Part 18: Sectional specification: Fixed aluminium electrolytic chip capacitors with solid and non-solid electrolyte
- Part 19: Sectional specification: Fixed metallized polyethylene-terephthalate film dielectric chip d.c. capacitors
- Part 20: Sectional specification: Fixed metallized polyphenylene sulfide film dielectric chip d.c. capacitors
- Part 21: Sectional specification: Fixed surface mount multilayer capacitors of ceramic dielectric, Class 1
- Part 22: Sectional specification: Fixed surface mount multilayer capacitors of ceramic dielectric, Class 2
- Part 23: Sectional specification: Fixed surface mount metallized polyethylene naphthalate film dielectric d.c. capacitors
- Part 24: Sectional specification – Surface mount fixed tantalum electrolytic capacitors with conductive polymer solid electrolyte (under consideration)
- Part 25: Sectional specification – Surface mount fixed aluminium electrolytic capacitors with conductive polymer solid electrolyte (under consideration)

All sectional specifications mentioned above do have one or more blank detail specifications being a supplementary document, containing requirements for style, layout and minimum content of detail specifications.

The QC 300401 number that appears on the front cover of this publication is the specification number in the IECQ Quality Assessment System for Electronic Components.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

FIXED CAPACITORS FOR USE IN ELECTRONIC EQUIPMENT –

Part 2-1: Blank detail specification: Fixed metallized polyethylene-terephthalate film dielectric d.c. capacitors – Assessment levels E and EZ

INTRODUCTION

Blank detail specification

A blank detail specification is a supplementary document to the sectional specification and contains requirements for style and layout and minimum content of details specifications. Detail specifications not complying with these requirements may not be considered as being in accordance with IEC specifications nor shall they be so described.

In the preparation of detail specifications the content of 1.4 of the sectional specification shall be taken into account.

The numbers between brackets on the first page correspond to the following information which shall be inserted in the position indicated:

Identification of the detail specification

- [1] The "International Electrotechnical commission" or the National Standards Organization under whose authority the detail specification is drafted.
- [2] The IEC or National Standards number of the detail specification, date of issue and any further information required by the national system.
- [3] The number and issue number of the IEC or national Generic Specification.
- [4] The IEC number of the blank detail specification.

Identification of the capacitor

- [5] A short description of the type of capacitor.
- [6] Information on typical construction (when applicable).
NOTE When the capacitor is not designed for use in printed board applications, this shall be clearly stated in the detail specification in this position.
- [7] Outline drawing with main dimensions which are of importance for interchangeability and/or reference to the national or international documents for outlines. Alternatively, this drawing may be given in an appendix to the detail specification.
- [8] Application or group of applications covered and/or assessment level.
NOTE The assessment level(s) to be used in a detail specification shall be selected from the sectional specification, 3.5.4. This implies that one blank detail specification may be used in combination with several assessment levels provided the grouping of the tests does not change.
- [9] Reference data on the most important properties, to allow comparison between the various capacitor types.

[1]	IEC 60384-2-1-XXX QC XXXXXXXXXXXX	[2]
ELECTRONIC COMPONENTS OF ASSESSED QUALITY IN ACCORDANCE WITH:	IEC 60384-2-1 QC XXXXXX	[4]
[3]	FIXED METALLIZED POLYETHYLENE-TEREPHTHALATE FILM DIELECTRIC DC CAPACITORS	[5]
Outline drawing (see table 1 (first angle projection)		[6]
[7]	Assessment levels E and EZ Performance grade:	[8]
(Other shapes are permitted within the dimensions given)		
NOTE For [1] to [9]: see previous page.		

Information on the availability of components qualified to this detail specification is given in IEC QC 001005

[9]

1 General data

1.1 Recommended method(s) of mounting (to be inserted)

See IEC 60384-2, 1.4.2.

1.2 Dimensions

Table 1 – Case size reference and dimensions

Case size reference	Dimensions						
	Ø	L	H	d			
When there is no case size reference, Table 1 may be omitted and the dimensions shall be given in Table 2, which then becomes Table 1.							
The dimensions shall be given as maximum dimensions or as nominal dimensions with a tolerance.							

1.3 Ratings and characteristics

Capacitance range (see Table 2)

Tolerance on rated capacitance

Rated voltage (see Table 2)

Category voltage (if applicable) (see Table 2)

Climatic category

Rated temperature

Maximum a.c. voltage (if applicable)

Maximum pulse load

Tangent of loss angle

Insulation resistance

Maximum dissipation

Table 2 – Values of capacitance and of voltage related to case sizes

Rated voltage				
Category voltage*				
	Case size	Case size	Case size	Case size
Rated capacitance (in nF and/or μ F)				
* If different from the rated voltage.				

1.4 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60384-1, *Fixed capacitors for use in electronic equipment – Part 1: Generic specification*

IEC 60384-2, *Fixed capacitors for use in electronic equipment – Part 2: Sectional specification: Fixed metallized polyethylene-terephthalate film dielectric d.c. capacitors*

1.5 Marking

The marking of the capacitor and the packing shall be in accordance with the requirements of IEC 60384-2, 1.6.

NOTE The details of the marking of the component and packing must be given in full in the detail specification.

1.6 Ordering information

Orders for capacitors covered by this specification shall contain, in clear or in coded form, the following minimum information:

- Rated capacitance;
- Tolerance on rated capacitance;
- Rated d.c. voltage;
- Number and issue reference of the detail specification and style reference.

1.7 Certified records of released lots

Required/not required.

1.8 Additional information (not for inspection purposes)

1.9 Additional or increased severities or requirements to those specified in the generic and/or sectional specification

NOTE Additions or increased requirements should be specified only when essential.

Table 3 – Other characteristics

This table is to be used for defining characteristics which are additional to or more severe than those given in the sectional specification.

2 Inspection requirements

2.1 Procedures

2.1.1 For qualification approval the procedures shall be in accordance with the sectional specification, IEC 60384-2, 3.4.

2.1.2 For quality conformance inspection the test schedule (Table 4) includes sampling, periodicity, severities and requirements. The formation of inspection lots is covered by 3.5.1 of the sectional specification IEC 60384-2.

Table 4 – Test schedule for Quality Conformance Inspection

Subclause number and test ^{a)}	D or ND ^{c)}	Conditions of test ¹⁾	IL	E lev.	EZ lev.		Performance requirements ^{a)}
				AQL ^{c)}	n	c	
Group A inspection (lot-by-lot) Subgroup A0 4.3.2 Capacitance 4.3.3 Tangent of loss angle 4.3.1 Voltage proof (Test A) 4.3.4 Insulation resistance (Test A)	ND	Frequency : 1 kHz for all capacitance values Method: ... Measuring point 1a Measuring point 1a			100% ^{d)}		Within specified tolerance As in 4.3.3.2 No breakdown or flashover. Self-healing breakdowns allowed As in 4.3.4.3
Subgroup A1 4.1 Visual examination 4.1 Dimensions (gauging)	ND		S-3	2,5%	^{b)}	0	As in 4.1 Legible marking and as specified in 1.5 of this specification As specified in Table 1 of this specification
Subgroup A2 4.2.2 Capacitance 4.2.3 Tangent of loss angle 4.2.1 Voltage proof (Test A) 4.2.4 Insulation resistance (Test A)	ND		S-3	1,0%	^{b)}	0	Within specified tolerance As in 4.2.3.2 No breakdown or flashover As in 4.2.4.2

Table 4 (continued)

Subclause number and test ^{a)}	D or ND ^{c)}	Conditions of test ¹⁾	IL	E lev.	EZ lev.		Performance requirements ^{a)}
				AQL ^{c)}	n	c	
Group B inspection (lot-by-lot) Subgroup B1 4.5 Solderability 4.15 Solvent resistance of the marking (if applicable)	ND	Without ageing Method: ... Solvent: ... Solvent temperature: ... Method 1 Rubbing material: ... Recovery time:	S-3	2,5%	^{b)}	0	Good tinning as evidenced by free flowing of the solder with wetting of the termina- tions or solder shall flow within ... s, as applicable Legible marking
NOTES: see end of table.							

Table 4 (continued)

Subclause number and test ^{a)}	D or ND ^{c)}	Conditions of test ^{a)}	Sample size and criterion of acceptability ^{c)}						Performance requirements ^{a)}
			E			EZ			
			<i>p</i>	<i>n</i>	<i>c</i>	<i>p</i>	<i>n</i>	<i>c</i>	
Group C inspection (periodic)									
Subgroup C1A Part of sample of Subgroup C1	D		6	9	1	6	5	0	
4.1 Dimensions (detail)									See detail specification
4.3.1 Initial measurements		Capacitance Tangent of loss angle: for $C_R > 1 \mu\text{F}$: at 1 kHz $C_R \leq 1 \mu\text{F}$: at 10 kHz							
4.3 Robustness of terminations		Visual examination							No visible damage
4.4 Resistance to soldering heat		Method:							
4.14 Component solvent resistance (if applicable)		Solvent: ... Solvent temperature: ... Method 2 Recovery time:							See detail specification
4.4.2 Final measurements		Visual examination Capacitance Tangent of loss angle							No visible damage Legible marking $\Delta C/C \leq 2 \%$ of value measured initially Increase of $\tan \delta$: $\leq 0,003$ $C \leq 1 \mu\text{F}$ Grade 1 $\leq 0,002$ $C > 1 \mu\text{F}$ Grade 1 $\leq 0,005$ $C \leq 1 \mu\text{F}$ Grade 2 $\leq 0,003$ $C > 1 \mu\text{F}$ Grade 2 compared to values measured in 4.3.1
Subgroup C1B Other part of sample of Subgroup C1	D		6	18	1	6	5	0	
4.6.1 Initial measurements		Capacitance Tangent of loss angle: for $C_R > 1 \mu\text{F}$: at 1 kHz $C_R \leq 1 \mu\text{F}$: at 10 kHz							
4.6 Rapid change of temperature		T_A = Lower category temperature T_B = Upper category temperature Five cycles Duration $t = 30$ min Visual examination							No visible damage
4.7 Vibration		Method of mounting: see 1.1 of this specification Frequency range: ... Hz to ... Hz Amplitude 0,75 mm or acceleration 100 m/s ² (whichever is the less severe) Total duration: 6 h							

Table 4 (continued)

Subclause number and test ^{a)}	D or ND ^{c)}	Conditions of test ^{a)}	Sample size and criterion of acceptability ^{c)}						Performance requirements ^{a)}
			E			EZ			
			<i>p</i>	<i>n</i>	<i>c</i>	<i>p</i>	<i>n</i>	<i>c</i>	
4.7.2 Final inspection		Visual examination							No visible damage
4.8 Bump (or shock, see 4.9)		Method of mounting: see 1.1 of this specification							
		Number of bumps: ... Acceleration: .. m/s ² Duration of pulse: ... ms							
4.9 Shock (or bump, see 4.8)		Method of mounting: see 1.1 of this specification							
		Acceleration: .. m/s ² Duration of pulse: ... ms							
4.8.3 or 4.9.3 Final measurements		Visual examination							No visible damage
		Capacitance							$\Delta C/C \leq 5\%$ of the value measured in 4.6.1
		Tangent of loss angle							Increase of $\tan \delta$: $\leq 0,003$ $C \leq 1\text{ }\mu\text{F}$ Grade 1 $\leq 0,002$ $C > 1\text{ }\mu\text{F}$ Grade 1 $\leq 0,005$ $C \leq 1\text{ }\mu\text{F}$ Grade 2 $\leq 0,003$ $C > 1\text{ }\mu\text{F}$ Grade 2 compared to values measured in 4.6.1
		Insulation resistance							$\geq 50\%$ of the values in 4.2.4.2
Subgroup C1 Combined sample of specimens of Subgroups C1A and C1B	D		6	27	2	6	10	0	
4.10 Climatic sequence									
4.10.2 Dry heat		Temperature: upper category temperature Duration: 16 h							
4.10.3 Damp heat, cyclic, Test Db, first cycle									
4.10.4 Cold		Temperature: lower category temperature Duration: 2 h							
4.10.5 Low air pressure (if required by the detail specification)		Air pressure: 8 kPa							
4.10.5.3 Intermediate measurement		Visual examination							No permanent breakdown, flashover or harmful deformation of the case
4.10.6 Damp heat, cyclic, Test dB, remaining cycles									
4.10.6.2 Final measurement		Visual examination							No visible damage Legible marking
		Capacitance							$\Delta C/C \leq 5\%$ of value measured in 4.4.2, 4.8.3 or 4.9.3 as applicable

Table 4 (continued)

Subclause number and test ^{a)}	D or ND ^{c)}	Conditions of test ^{a)}	Sample size and criterion of acceptability ^{c)}						Performance requirements ^{a)}
			E			EZ			
			<i>p</i>	<i>n</i>	<i>c</i>	<i>p</i>	<i>n</i>	<i>c</i>	
		Tangent of loss angle							Increase of tan δ: ≤0,005 <i>C</i> ≤ 1 μF Grade 1 ≤0,003 <i>C</i> > 1 μF Grade 1 ≤0,008 <i>C</i> ≤ 1 μF Grade 2 ≤0,005 <i>C</i> > 1 μF Grade 2 compared to values measured in 4.3.1 or 4.6.1 as applicable
		Insulation resistance							≥50 % of values in 4.2.4.2
Subgroup C2	D		6	15	1	6	10	0	
4.11 Damp heat, steady state									
4.11.1 Initial measurements		Capacitance Tangent of loss angle at 1 kHz							No visible damage Legible marking
4.11.3 Final measurement		Visual examination							Δ <i>C</i> / <i>C</i> ≤ 5 % of the value measured in 4.11.1
		Capacitance Tangent of loss angle							Increase of tan δ: ≤0,005 compared to the values measured in 4.11.1
		Insulation resistance							≥50 % of values in 4.2.4.2
Subgroup C3	D		3	21	1	6	10	0	
4.12 Endurance		Duration: Grade 1: 2 000 h Grade 2: 1 000 h							
4.12.1 Initial measurement		Capacitance Tangent of loss angle: for <i>C_R</i> > 1 μF: at 1 kHz <i>C_R</i> ≤ 1 μF: at 10 kHz							No visible damage Legible marking
4.12.5 Final measurement		Visual examination							Δ <i>C</i> / <i>C</i> ≤ 5 % for Grade 1 ≤8 % for Grade 2 of value measured in 4.12.1
		Capacitance							Increase of tan δ: ≤0,003 <i>C</i> ≤ 1 μF Grade 1 ≤0,002 <i>C</i> > 1 μF Grade 1 ≤0,005 <i>C</i> ≤ 1 μF Grade 2 ≤0,003 <i>C</i> > 1 μF Grade 2 compared to values measured in 4.12.1
		Tangent of loss angle							≥50 % of values in 4.2.4.2
		Insulation resistance							≥50 % of values in 4.2.4.2

Table 4 (continued)

Subclause number and test a)	D or ND c)	Conditions of test a)	Sample size and criterion of acceptability c)						Performance requirements a)
			E			EZ			
			p	n	c	p	n	c	
Subgroup C4	D		3	9	1	6	10	0	
4.13 Charge and discharge									
4.13.1 Initial measurement		Capacitance Tangent of loss angle: for C _R > 1 μF: at 1 kHz C _R ≤ 1 μF: at 10 kHz Duration of charge: ... s Duration of discharge: ... s							
4.13.3 Final measurement		Capacitance Tangent of loss angle Insulation resistance							ΔC/C ≤3 % for Grade 1 ≤5 % for Grade 2 of value measured in 4.13.1 Increase of tan δ: ≤0,003 C ≤ 1 μF Grade 1 ≤0,002 C > 1 μF Grade 1 ≤0,005 C ≤ 1 μF Grade 2 ≤0,003 C > 1 μF Grade 2 compared to values measured in 4.13.1 ≥50 % of values in 4.2.4.2
a) Subclause numbers of tests and performance requirements refer to the sectional specification, IEC 60384-2 and clause 1 of this specification. b) Number to be tested: Sample size as directly allotted to the code letter for IL in Table 2a of IEC 60410 (Single sampling plan for normal inspection) c) In this table: p = periodicity (in months); n = sample size; c = acceptance criterion (permitted number of defectives); D = destructive; ND = non-destructive; IL = inspection level; AQL = acceptable quality level } IEC 60410 d) 100% testing shall be followed by re-inspection by sampling in order to monitor outgoing quality level by non-conforming items per million (ppm). The sampling level shall be established by the manufacturer. For the calculation of ppm values any parametric failure shall be counted as non-conforming item. In case one or more non-conforming items occur in a sample, this lot shall be rejected.									



Standards Survey

The IEC would like to offer you the best quality standards possible. To make sure that we continue to meet your needs, your feedback is essential. Would you please take a minute to answer the questions overleaf and fax them to us at +41 22 919 03 00 or mail them to the address below. Thank you!

Customer Service Centre (CSC)

International Electrotechnical Commission

3, rue de Varembe
1211 Genève 20
Switzerland

or

Fax to: **IEC/CSC** at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

A Prioritaire

Nicht frankieren
Ne pas affranchir



Non affrancare
No stamp required

RÉPONSE PAYÉE

SUISSE

Customer Service Centre (CSC)
International Electrotechnical Commission
3, rue de Varembe
1211 GENEVA 20
Switzerland



Q1 Please report on **ONE STANDARD** and **ONE STANDARD ONLY**. Enter the exact number of the standard: (e.g. 60601-1-1)

.....

Q2 Please tell us in what capacity(ies) you bought the standard (tick all that apply). I am the/a:

- purchasing agent ☐
 librarian ☐
 researcher ☐
 design engineer ☐
 safety engineer ☐
 testing engineer ☐
 marketing specialist ☐
 other.....

Q3 I work for/in/as a:
(tick all that apply)

- manufacturing ☐
 consultant ☐
 government ☐
 test/certification facility ☐
 public utility ☐
 education ☐
 military ☐
 other.....

Q4 This standard will be used for:
(tick all that apply)

- general reference ☐
 product research ☐
 product design/development ☐
 specifications ☐
 tenders ☐
 quality assessment ☐
 certification ☐
 technical documentation ☐
 thesis ☐
 manufacturing ☐
 other.....

Q5 This standard meets my needs:
(tick one)

- not at all ☐
 nearly ☐
 fairly well ☐
 exactly ☐

Q6 If you ticked NOT AT ALL in Question 5 the reason is: (tick all that apply)

- standard is out of date ☐
 standard is incomplete ☐
 standard is too academic ☐
 standard is too superficial ☐
 title is misleading ☐
 I made the wrong choice ☐
 other

Q7 Please assess the standard in the following categories, using the numbers:

- (1) unacceptable,
 (2) below average,
 (3) average,
 (4) above average,
 (5) exceptional,
 (6) not applicable

- timeliness.....
 quality of writing.....
 technical contents.....
 logic of arrangement of contents
 tables, charts, graphs, figures.....
 other

Q8 I read/use the: (tick one)

- French text only ☐
 English text only ☐
 both English and French texts ☐

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

.....



ISBN 2-8318-8335-0



ICS 31.060.30
